ABHIJITH BALACHANDRAN

Halifax, Nova Scotia, Canada | +1-(647)-883-6282 | abhijith.uwc.rw@gmail.com | Linkedin

Embedded Software Engineer with 8+ years of experience developing embedded systems in C/C++ for microcontroller- and Linux-based platforms. Experienced in firmware development, device drivers, RTOS (FreeRTOS), Embedded Linux, bare-metal programming, and hardware debugging using oscilloscopes and logic analyzers. Proficient in communication protocols including UART, SPI, I2C, CAN, and TCP/IP, with a strong focus on writing reliable, well-documented code and delivering high-quality embedded software in collaborative engineering environments.

EXPERIENCE

BinSentry **03/2025 to 03/2026**

Senior Embedded Software Developer

- Developed and maintained IoT firmware using FreeRTOS on ARM Cortex-based STM32 microcontrollers, integrating wireless connectivity with Morse Micro HaLow (802.11ah) technology.
- Utilized C/C++, FreeRTOS APIs, STM32CubeIDE, AWS IoT Core, and Serverless Framework for end-to-end IoT solution development and integration.
- Performed code reviews and contributed to continuous improvement of firmware architecture and coding standards.
- Designed and integrated AWS Cloud connectivity using MQTT for secure IoT data communication between devices and cloud services.
- Deployed a Linux-based AWS environment with a Kubernetes cluster to host OpenWISP, enabling scalable, containerized network management and gaining hands-on experience with cloud-native orchestration and Linux system administration.

Microchip Technology **06/2022 to 12/2024**

Senior Software Engineer

- Developed multi-threaded embedded Linux applications and device drivers in C/C++ using POSIX APIs, with a focus on networking, hardware interfacing, and system performance on embedded platforms.
- Developed BSPs for embedded switch platforms using Linux kernel, U-Boot, Device Tree (DTS), and Yocto/Buildroot. Extended Linux device drivers (I2C, SPI, GPIO, PCIe) and developed firmware for SFP, SFP+, and OSFP transceivers, supporting board bring-up and hardware validation.x projects.
- Implemented CI/CD pipelines using Bitbucket and Jenkins with integrated build and test automation, static code analysis using Coverity, and build orchestration with CMake, enabling streamlined delivery and improved code quality across embedded Linux Projects.
- Led the development and open-source release of an OS-agnostic user-space Ethernet PHY driver CMake project, incorporating SDLC best practices. It supports Serdes, ANEG, IEEE 1588, MDIO, and MACsec, for Copper/Optical PHYs (OTN) on both FPGA and ASIC platforms.
- Developed a test framework in Python and Ruby for automated API testing (unit, functional and regression), collaborating with cross-functional teams to ensure comprehensive test coverage and seamless integration. Utilized iPerf to measure throughput and performance.
- Mentored junior engineers by providing guidance, sharing industry best practices, and leading hands-on sessions and code reviews.

Gadgeon Smart Systems Pvt Ltd **05/2021 to 06/2022**
Senior Software Engineer

- Led a team for developing Linux device drivers for the camera module based on V4L framework and peripherals(I2C, DMA, GPIO, ISC).
- Managed embedded switch applications using TCP/IP, VLAN, OSPF, BGP, MPLS and DPDK in C/C++, implementing new features, resolving issues, and developing Python test scripts for updates. Delivered thorough documentation for bug fixes and improvements.
- Worked extensively with Yocto and Buildroot to add custom Rootfs packages, implement secure boot, and apply kernel patches for switch platforms. Also mentored junior engineers on these technologies to help enhance their skills and knowledge.
- Engaged in hardware debugging using an oscilloscope and logic analyzer, and developed device trees based on hardware schematics for peripherals like I2C, SPI, ADCs, DACs, Clocks, and UART.

NPOL DRDO **12/2019 to 05/2021**
Research Fellow

- Carried out research in underwater communication, channel modeling, and benchmarking, with the results published in IEEE journals.
- Developed digital signal processing algorithms and MATLAB simulations, along with C/C++ software, for underwater acoustic modems.
- Mentored junior engineers and participated in code reviews, fostering knowledge sharing and ensuring code quality

Resnova Technologies **07/2017 to 07/2019**
Software Developer

- Gained hands-on experience in embedded programming (C/C++) and build systems (Yocto, Buildroot), contributing to firmware stability and efficiency.
- Developed an IoT gateway on a Linux platform to collect data from a Zigbee network and transmit it securely to cloud services, implementing efficient data processing, protocol handling, and integration with MQTT-based cloud endpoint

SKILLS

• C/C++ • Zephyr OS • FreeRTOS • Linux Kernel/Device Drivers • ARM Cortex • Networking protocols • CMake • Yocto • Buildroot • OpenWRT • Python • Bash • Git • I2C • SPI • UART • GPIO • MQTT • WiFi

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering **08/2013 to 06/2017**
University of calicut

PUBLICATIONS & CERTIFICATIONS

- IEEE: Chirp Slope Keying: A Practical Benchmark Modulation Scheme for Underwater Acoustic Channel Replay Simulation
- IEEE: Impact of Doppler on narrowband and wideband signals
- Linux Certifications: Linux Kernel Programming, Memory Management in Linux,Yocto Embedded Linux